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Dear Applicant,

There is less than 48 hours remaining to apply for the 2027 NATO DIANA Programme.



The submission window will close on **3 July 2026 at 12:00 BST/UTC +1**.

Please remember to compete and submit your submission before this time - any proposals still in draft at the time of closure will not be considered for evaluation.

Withdraw

Confidential

Status:

Submitted

Hybrid propulsion system VTOL drone with tilt rotor system for medical and defence applications capable of long-range deliveries

TRL LEVEL

Please select the Technology Readiness Level ("TRL") of the technical solution you are proposing to the challenge.

TRL 4 Technology validated in lab

TECHNICAL TAGS

AI, Autonomy & Decision Systems > Situational Awareness

Unmanned Systems (Air, Land, Sea) > Unmanned Systems

Unmanned Systems (Air, Land, Sea) > Drone UAV

Unmanned Systems (Air, Land, Sea) > UAS Uxs

Unmanned Systems (Air, Land, Sea) > Autonomous Navigation

Unmanned Systems (Air, Land, Sea) > GPS Denied Navigation

Unmanned Systems (Air, Land, Sea) > Path Planning

Unmanned Systems (Air, Land, Sea) > Obstacle Detection

Unmanned Systems (Air, Land, Sea) > SLAM Mapping

Unmanned Systems (Air, Land, Sea) > Autonomous Control

Unmanned Systems (Air, Land, Sea) > Mission Autonomy

Unmanned Systems (Air, Land, Sea) > Payload Carrier Platform

Medical, Human Factors & Physiology > Medical Monitoring

Medical, Human Factors & Physiology > Triage Systems

Medical, Human Factors & Physiology > Biomedical Carrier System

SYSTEM TYPE

State whether your technical solution is software, hardware or a combination of both.

Hardware



System

Withdraw

PART 2 - TECHNICAL APPROACH

- Describe your technical solution. You may attach graphics or images to help convey your concept.

1. Full-scale concept:

- The full-scale OrgoFly concept uses a tilt-rotor VTOL architecture combining runway-independent access with efficient fixed-wing cruise for medical logistics.
- For take-off and landing, rotors point vertically to generate lift from confined or improvised areas.
- After climb-out, rotors rotate forward, and the aircraft transitions into wing-borne cruise.

2. Airframe and propulsion:

- Unlike VTOL UAVs with separate lift rotors and a pusher motor, OrgoFly uses the same rotors for hover and forward flight.
- This adds complexity, but reduces redundant propulsion hardware and improves aerodynamic efficiency, range, and mission flexibility.
- The full-scale concept includes a folding wing for transport, storage, and deployment from constrained environments.
- Hybrid-electric propulsion provides endurance, energy efficiency, and operational resilience.
- A redundant battery system supports emergency propulsion and auxiliary equipment.

3. Medical payload system:

- The full-scale concept includes a temperature-controlled medical payload compartment for safe medical transport.
- Payload conditions are continuously monitored to support traceability and usability.

4. Operational capability:

- Together, these elements enable runway-independent delivery from field hospitals, remote bases, ships, islands, disaster zones, and improvised landing areas.
- Target full-scale capability is 15 kg payload and up to 400 km range, enabling meaningful medical logistics beyond small package delivery

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PART 3 - TECHNOLOGY DEVELOPMENT ROADMAP WITH RISKS AND MITIGATIONS

- How will the challenge funding help you to mature your proposed technical solution into a demonstrable and/or deployable technology?
- Key milestones:
 - Six months: What do you plan to achieve in six months?
 - One year: If selected for Mission Track and further challenge funding were offered, what would you intend to achieve in one year?
- Risks: What are the most significant risks associated with achieving these milestones?
- Mitigation: What measures are you proposing to mitigate these risks?

1. Funding use:

- DIANA funding supports the 1-year phase after the bootstrapped 1:4 concept.
- Used for full-scale prototype and certification: folding wing, hybrid propulsion, modular payload bay, temperature-controlled bay, avionics, testing, and BVLOS prep.

2. Six months (bootstrapped):

- Demonstrate full flight capabilities in all operational phases and environments.
- Gather data on stability, range, payload effects, reliability, and scaling.
- Demonstrate successful downsized payload delivery.
- Initiate certification process: safety case, telemetry, failsafes, logs.

3. One year: (with funding)

- Build full-scale prototype with hybrid propulsion, folding wing, modular, temperature-controlled cargo bay.
- Demonstrate full defence profile.
- Progress BVLOS certification evidence.

4. Risks:

- Civilian certification delays - BVLOS, Specified category, medical.
- EW/GNSS degradation, contested airspace, kinetic threats including interceptor drones.
- Evolving EASA/FAA civilian UAV airspace use regulations.
- Rapid battlefield evolution may shift mission needs..

5. Mitigations:

- Conduct permitted civilian operations VLOS and focus on defence mission profile.
- Build certification evidence early through tests, logs, failsafes, and regulator/test-site engagement.
- Integrate state-of-the-art navigation and communication components that are modular and upgradeable.
- Use modular, upgradable navigation, communications, propulsion, and payload architecture.
- Update mission profiles through user/defence feedback.

PART 4 - BUSINESS CASE

- What are the applications of your solution in the defence and security sector? Describe any routes to adoption that you have identified.
- What is the market for your solution?
- What is your current level of investment from public and/or private sources?
- Do you have existing relationships with industry primes? Do you intend to partner with a systems integrator or provide an integrated solution?



Details

Withdraw...

1. Defence/security applications:

- Forward medical resupply to dispersed units in contested ground/airspace.
- Delivery of trauma kits, medicines, diagnostics, countermeasures, blood, plasma, and warming equipment.
- Support to prolonged field care when evacuation is delayed.
- Resilient logistics where convoys, helicopters, runways, or large hubs are unavailable.

2. Market:

- Civilian beachhead: hospital-to-hospital organ and sensitive medical delivery without dedicated infrastructure.
- Organ logistics is a niche but \$600M market, with strong fit covering 80% of missions with our 15 kg / 400 km organ-box.
- Expansion path into transplant-adjacent urgent freight: samples, blood, plasma, diagnostics, and medical consumables.

3. Routes to adoption:

- Use civilian applications to monetize early operations and support validation and certification efforts alongside defence-related development.
- Defence route through GATE Institute – A Partner for Innovation in Security and Defence.
- DIANA would support user validation, testing, and access to defence/security stakeholders.

4. Investment:

- Current development is bootstrapped toward the 1:4 concept.
- DIANA funding would accelerate full-scale prototype development and the certification process.

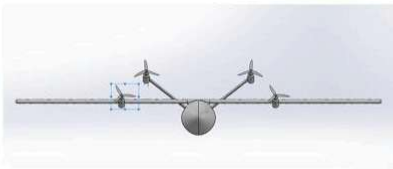
5. Partnerships:

- GATE Institute is the identified defence/security innovation partner.
- We intend to provide an integrated solution and partner with Tier 1 and Tier 2 defence contractors to tailor to the defence segment

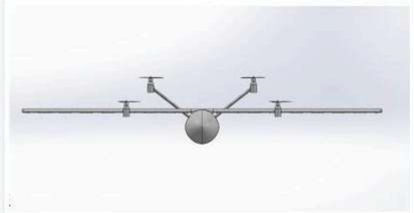
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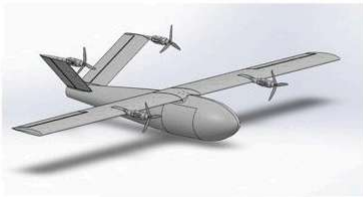
Front Motor Forward Flight



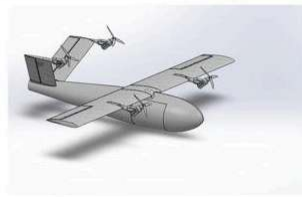
Front Motor Transition



Front Motor VTOL


Details

Side Left Motor Forward Flight

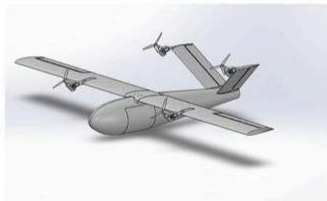
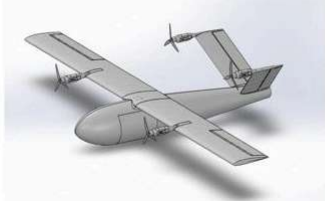


Side Left Motor Transition

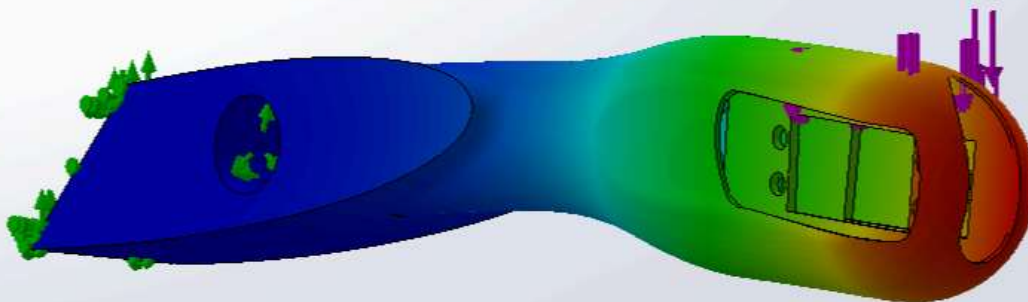


Side Left Motor VTOL

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Model name: Lower_Fairing_v3
Study name: Hover_Static_MAX(-Default-)
Plot type: Static displacement Displacement1
Deformation scale: 11,311



Long form proposal

EMERGING AND DISRUPTIVE TECHNOLOGIES

Autonomous Systems

ADDITIONAL TAGS

VTOL, Hybrid, Tilt-rotor, temp-controlled cargo

NATO DIANA 2026

SUPPORT

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Terms and Conditions FAQs



Details



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